

Occupation and risk of lymphoma: a multicentre prospective cohort study (EPIC).



OBJECTIVES: Evidence suggests that certain occupations and related exposures may increase the risk of malignant lymphoma. Farming, printing and paper industry, wood processing, meat handling and processing, welding, shoe and leather manufacturing and teaching profession are among the categories that have been implicated in previous studies. The relationship between occupation and malignant lymphoma has been investigated in a large European prospective study. **METHODS:** We investigated occupational risks for lymphomas in the European Prospective Investigation into Cancer and Nutrition (EPIC). The mean follow-up time for 348,555 subjects was 9 years (SD: 2 years). The analysis was based on 866 and 48 newly diagnosed cases of non-Hodgkin's lymphoma (NHL) and Hodgkin's lymphoma (HL). These were identified in the EPIC subcohorts with occupational data. Data on 52 occupations were collected through standardised questionnaires. Cox proportional hazard models were used to explore the association between occupation and risk of malignant lymphoma. **RESULTS:** The following occupations were positively associated with malignant NHL after adjustment for study centre, age, sex, socioeconomic status (SES), smoking and alcohol: butchers (HR=1.53, 95% CI 1.05 to 2.48, including multiple myeloma/plasmacytoma; HR=1.30, 95% CI 1.00 to 2.66, excluding multiple myeloma/plasmacytoma) and car repair workers (HR=1.50, 95% CI 1.01 to 2.00, including multiple myeloma/plasmacytoma; HR=1.51, 95% CI 1.01 to 2.31, excluding multiple myeloma/plasmacytoma). HL was associated with gasoline station occupation (HR=4.59, 95% CI 1.08 to 19.6). **CONCLUSION:** The findings in this current study of a higher risk of NHL among car repair workers and butchers and a higher risk of HL among gasoline station workers suggest a possible role from occupationally related exposures, such as solvents and zoonotic viruses, as risk factors for malignant lymphoma.